KPPSC PMS Geography Paper-I — Q.6: Glacial Geomorphology: Valley Glaciers vs. Ice Sheets

KPPSC PMS Master Examination Notes — Paper-I (Physical & General Geography)

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-I (Physical & General Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• **2022 Q.6:** "Discuss valley glaciers and the resultant erosional and depositional landforms in detail. (20 Marks)" • **2018 Q.8(a):** "Write short note on: Glacial landforms. (10 Marks)" • **2010 Q.3:** "Differentiate between Valley Glacier and Ice Sheet. Discuss the depositional landforms of Valley Glacier in detail. (20 Marks)"
Target Syllabus	Section 2: Landforms — Glacial Landforms: Valley Glaciers vs. Ice Sheets
Marks & Allocation	20 Marks | Strict 35-Minute Descriptive Writing Limit
Core Theoretical Thesis	Glacial ice operates as a colossal plastic-solid geomorphic engine driven by cryospheric compaction, basal sliding, and internal deformation. Glaciers divide fundamentally into topographically confined Valley (Alpine) Glaciers and unconfined continental Ice Sheets. Operating via plucking and glacial abrasion, alpine glaciers transform pre-existing fluvial V-shaped valleys into steep-walled, flat-bottomed U-shaped glacial troughs flanked by hanging valleys, cirques, and arêtes, while depositional mechanics generate unsorted moraines, drumlins, and outwash plains—typified globally by the Karakoram giants (Siachen, Baltoro, Batura) of northern Pakistan.

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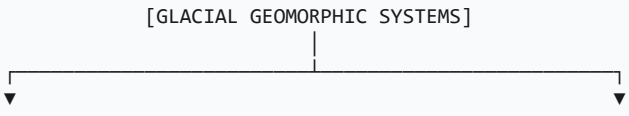
CRYOSPHERIC ICE DYNAMICS: Snow transforms into glacial ice through compaction and recrystallization: Fresh Snow (density 0.1 g/cm^3) -> Granular Firn/Névé (0.4–0.8 g/cm^3) -> True Glacial Ice (>0.85 g/cm^3). Glacial ice flows under gravity once ice thickness exceeds ~30–50 meters via internal plastic deformation (creep) and basal sliding.

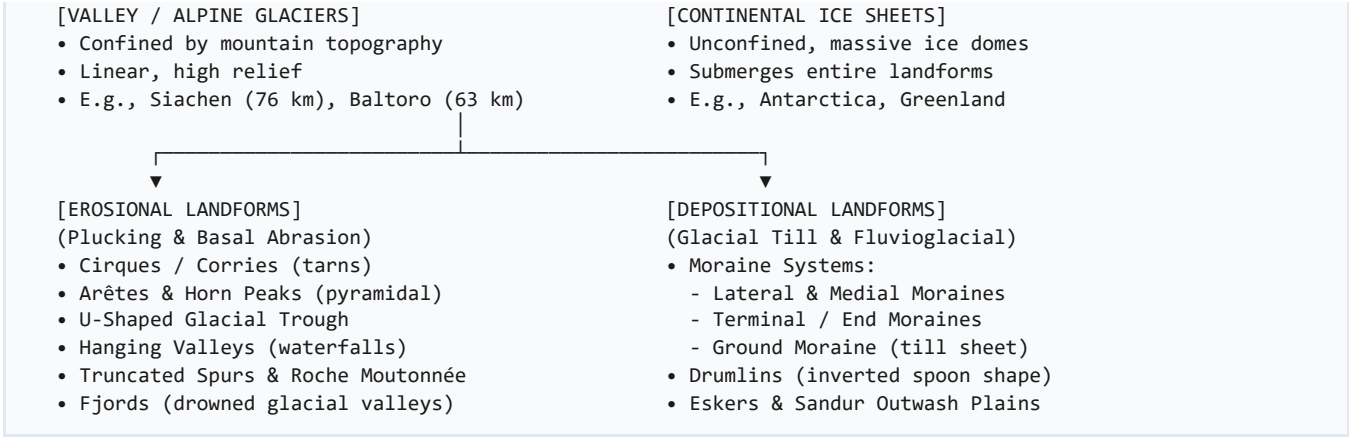
VALLEY GLACIERS VS. ICE SHEETS DICHOTOMY: Valley/Alpine Glaciers are linear ice tongues confined by pre-existing mountain valley walls (e.g., Baltoro, Siachen, Mer de Glace); Continental Ice Sheets are unconfined, dome-shaped continental ice masses submerging topography under kilometers of ice (e.g., Antarctica: 14M km^2, Greenland: 1.7M km^2).

EROSIONAL MASTER FEATURES: (1) Cirque/Corrie (horseshoe amphitheater hollow excavated by freeze-thaw plucking and rotational slip); (2) Arête (knife-edge ridge separating two cirques); (3) Horn Peak (pyramidal peak formed by 3+ intersecting cirques, e.g., Matterhorn, K2); (4) U-Shaped Trough (steep-sided, flat-floored glacial valley); (5) Hanging Valleys (tributary valleys left suspended above over-deepened main trough, forming waterfalls); (6) Fjords (submerged glacial troughs).

DEPOSITIONAL ARCHITECTURE: Glacial Till / Moraines (unsorted, unstratified direct glacial debris): Lateral, Medial, Ground, and Terminal/Recessional Moraines; Fluvioglacial stratified deposits: Drumlins (streamlined teardrop till mounds), Eskers (sinuous subglacial meltwater river gravel ridges), Kames, and Outwash Plains (sandurs).

1. Conceptual Mind Map & Thematic Roadmap





2. Analytical Dimensions & Geomorphic/Systemic Matrix

Glacial Landform Class	Geomorphic Genesis & Mechanism	Diagnostic Diagnostic Examples & Pakistan Himalayan Context
Valley Glaciers vs. Continental Ice Sheets	Valley glaciers are constrained by mountain topography; continental sheets are unconfined, radial ice domes.	Valley: Siachen (76 km), Biafo (67 km), Baltoro (63 km), Batura (57 km) in Karakoram. Ice Sheet: Antarctic Ice Sheet (>4,000 m thick).
Cirque (Corrie / Cwm)	Rotational slip of ice combined with frost plucking at bergschrund headwall; carves armchair hollow.	Deep amphitheater-shaped rock basin; when ice melts, occupied by a circular alpine tarn lake (e.g., Saif-ul-Malook tarn in Kaghan Valley).
Arêtes and Horn Peaks	Backward recession of opposing cirque headwalls sharpens intervening mountain ridges and summits.	Knife-edge saw-tooth arêtes; iconic pyramidal Horn peaks (e.g., Matterhorn in Alps; K2, Broad Peak in Karakoram Range).
U-Shaped Glacial Trough	Glacier widens, deepens, and straightens pre-existing fluvial V-shaped valley via plucking and abrasion.	Steep, vertical valley walls, broad flat valley floor, truncated spurs, ribbon lakes (e.g., Swat, Hunza, and Gilgit alpine valleys).
Hanging Valleys	Main valley glacier incises far deeper than smaller tributary glaciers due to greater ice volume and erosive mass.	Tributary valleys suspended high above main valley floor; tributaries cascade down as spectacular waterfalls (e.g., Hunza Valley tributaries).
Moraine Architecture (Glacial Till)	Direct, unstratified, unsorted dump of angular rock debris transported supraglacially and subglacially.	Lateral moraines along valley flanks, medial moraines where two ice tongues fuse, and terminal end moraines damming glacial valleys.

3. 20-Mark Master Model Essay

Introduction: The Cryosphere as a Geomorphic Force

Glaciers constitute one of the most dynamic and powerful physical agents of subaerial denudation on Earth. Today, glacial ice blankets approximately 10 percent of the planet's subaerial land surface—predominantly locked within the colossal polar ice sheets of Antarctica and Greenland—while during the Pleistocene Epoch (the Quaternary Ice Age), glaciers expanded across more than 30 percent of continental landmasses. Outside the polar regions, the greatest concentration of glacial ice resides in the towering Karakoram, Himalayan, and Hindu Kush mountain ranges of northern Pakistan, rightly designated as Earth's 'Third Pole'. Glaciers originate wherever annual snowfall exceeds summer melting (ablation), causing fresh snow (density 0.1 g/cm^3) to compact under successive layers into granular firn (or névé, 0.4–0.8 g/cm^3) and ultimately recrystallize into dense, impermeable glacial ice (>0.85 g/cm^3). Once ice thickness reaches a threshold of 30 to 50 meters, the shear stress generated by gravity surpasses the yield strength of ice, causing the glacier to deform internally like a hyper-viscous plastic solid and surge downslope.

Valley Glaciers vs. Continental Ice Sheets: A Morphological Dichotomy

In physical geography, glaciers are categorized into two fundamental structural classes: Valley (Alpine) Glaciers and Continental Ice Sheets. Valley glaciers are linear tongues of ice that originate in high-altitude mountain snowfields (cirques) and flow down pre-existing stream valleys, strictly confined by surrounding bedrock topography. Their

movement is highly channelized, their gradients are steep, and their morphological work is focused intensely upon valley deepening and relief accentuation. Pakistan hosts the world's longest alpine glaciers outside the polar circles, including the Siachen Glacier (76 km), the Biafo Glacier (67 km), the Baltoro Glacier (63 km), and the Batura Glacier (57 km). In stark contrast, Continental Ice Sheets (and Ice Caps) are unconfined, gargantuan domes of ice that submerge entire continental landmasses under ice sheets 2 to 4 kilometers thick. Their flow is radial, moving outward from elevated central domes toward marine margins regardless of underlying bedrock topography, as observed in the 14-million-square-kilometer Antarctic Ice Sheet. Whereas valley glaciers sharpen and steepen alpine peaks into jagged relief, continental ice sheets act as regional planes, grinding down continental shields into undulating, low-relief roche moutonnée landscapes.

Mechanics of Glacial Erosion: Plucking, Abrasion, and Subglacial Cavitation

Glacial erosion operates primarily through two complementary mechanical processes: Plucking (quarrying) and Abrasion. 'Plucking' occurs along the fractured, jointed bedrock of the glacier's bed. Subglacial meltwater percolates into bedrock fissures, where pressure variations cause it to freeze, shattering rock blocks through frost wedging. As the overlying glacier advances, these shattered bedrock fragments are frozen into the basal ice and physically plucked from the bed, incorporated into the moving glacier as subglacial tools. 'Abrasion' is the subsequent rasping, scouring, and grooving of the bedrock bed by these embedded, angular rock fragments, functioning like coarse industrial sandpaper. Abrasion polishes bedrock surfaces to a mirror-like sheen, carves parallel linear 'glacial striations' that reveal past ice flow directions, and pulverizes rock into microscopic 'rock flour', which imparts a distinctive milky turquoise tint to downstream glacial meltwater streams.

Erosional Alpine Landforms: Cirques, Arêtes, Horns, and U-Shaped Troughs

Valley glaciation completely restructures mountain topography. The foundational erosional feature is the 'Cirque' (or Corrie in Scotland, Cwm in Wales)—a steep-walled, armchair-shaped amphitheater hollow carved into the mountain headwall. Rotational slip of the ice mass combined with freeze-thaw plucking along the deep headwall crevasse (the bergschrund) steepens the backwall while scooping out a rock basin floor. When the glacier melts, this over-deepened basin frequently fills with water to form an enchanting circular alpine lake known as a 'Tarn' (such as Lake Saif-ul-Malook in Pakistan's Kaghan Valley). When two adjacent cirques erode backward toward each other, their intersecting headwalls sharpen the intervening ridge into a razor-thin, knife-edge serrated ridge termed an 'Arête'. Where three or more cirques erode radially into a single mountain summit, they quarry away all flanks to leave an isolated, towering pyramidal peak known as a 'Horn'—immortalized globally by the Swiss Matterhorn and regionally by the monumental pyramids of K2 and Broad Peak in the Karakoram. As the valley glacier bulldozes down-valley, it truncates pre-existing river spurs, widening, deepening, and straightening the valley into a monumental 'U-Shaped Glacial Trough' characterized by sheer vertical valley walls and a flat, broad floor. Furthermore, because tributary glaciers contain far less ice volume than the trunk glacier, they incise their beds far less deeply; when the ice vanishes, tributary valleys are left suspended high above the main trough floor as 'Hanging Valleys', from which meltwater streams plunge as spectacular waterfalls.

Depositional Architecture: Moraines, Drumlins, Eskers, and Outwash Plains

Glacial deposition produces two distinct genetic suites of sediments: unstratified glacial till (moraine) dropped directly by melting ice, and stratified fluvioglacial drift reworked and sorted by meltwater streams. The primary depositional landform is the 'Moraine'—an unsorted, unstratified ridge of angular boulders, gravel, sand, and clay. 'Lateral Moraines' accumulate along the margins of valley glaciers from frost-shattered valley rockfall; where two tributary glaciers merge, their adjacent lateral moraines coalesce to form a prominent dark ribbon of debris running down the center of the trunk glacier, known as a 'Medial Moraine' (vividly visible along the Baltoro Glacier). At the maximum snout advance where ablation equals ice supply, debris is dumped to form a massive crescentic 'Terminal End Moraine', which frequently impounds meltwater to create proglacial lakes. Fluvioglacial landforms include 'Drumlins'—streamlined, teardrop-shaped hills of compacted till shaped like inverted spoons, with their steep blunt ends facing the direction of up-glacier advance. Subglacial meltwater streams running in pressurized sub-ice tunnels deposit sinuous, gravel ridges known as 'Eskers', while braided meltwater streams emerging from the glacier snout deposit expansive, sorted sand and gravel plains termed 'Outwash Plains' (or Sandurs).

Conclusion: The Glacial Engine and Pakistan's Hydrological Security

In evaluative summation, glacial geomorphology illustrates the supreme denudational capacity of the solid cryosphere. Glaciers act as natural bulldozers, transforming rugged fluvial terrains into dramatic, high-energy glacial landscapes that

define the alpine world. Beyond their geomorphic splendor, glaciers serve as the irreplaceable freshwater towers of humanity. In Pakistan, the extensive glaciated massifs of the Karakoram, Himalayas, and Hindu Kush hold over 7,000 individual glaciers containing thousands of cubic kilometers of frozen freshwater. These glaciers feed the Indus River Basin, sustaining over 80 percent of Pakistan's agricultural irrigation, hydroelectric generation, and urban water supply. Today, anthropogenic climate change is accelerating glacial retreat, increasing the frequency of Glacial Lake Outburst Floods (GLOFs) across Chitral, Gilgit, and Hunza, and threatening the long-term hydrological stability of South Asia. Understanding glacial processes is thus not merely an academic exercise, but a strategic imperatve for Pakistan's national survival.

4. Scholarly Perspectives & Geomorphic Axioms

Scholar / Glaciologist	Theoretical Contribution / Concept	Exact Quotable Academic Thesis / Axiom
Louis Agassiz (1840)	Études sur les Glaciers (The Glacial Theory)	"*The polished bedrock, striations, and erratic boulders across northern continents are not the work of a biblical deluge, but the undeniable signature of colossal vanished ice sheets.*"
J.T. Andrews (1975)	Glacial Systems: An Approach to Glaciers and Their Landforms	"*A glacier is a thermodynamic engine operating in open mass-balance equilibrium; its morphology is a direct response to inputs of snow accumulation and outputs of ablation.*"
John F. Nye (1952)	The Mechanics of Glacier Flow	"*Glacial ice behaves as a non-linear viscoplastic material; when shear stress at the bed exceeds a critical yield threshold, continuous plastic flow and basal sliding occur.*"
Clifford Embleton & C.A.M. King (1975)	Glacial Geomorphology	"*The conversion of a fluvial V-shaped valley into a glacial U-shaped trough represents one of the most complete geomorphic transformations in physical geography.*"
Kenneth Hewitt (2005)	The Karakoram Anomaly & Glacial Dynamics	"*The Karakoram cryosphere exhibits exceptional behavior, where high-altitude surging glaciers and debris-covered ablation zones maintain a delicate, resilient equilibrium amidst global warming trends.*"